ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

NCIOP 00114

Submarine Naval Combat Information Operator

NQUAL - AILD

MITE IDs
Module 1 (DLN) - 123106
Module 2 (Residential) - 123107
(OJT) - 109269

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Center Atlantic (NTDC(A)) is the designated development center responsible for the maintenance of training. Amendments to PO standards are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
V1.0	21 April 2007	DMTE
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		CNPTG

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FOREWORD

1. This training is based on the specification found at:

   a. A-P9-050 Series, Canadian Forces Individual Training and Education System Manuals;

   b. A-PD-055-002/PP-02 Occupation Specification (NCIOP) Dated 14 Jan 2016; and

   c. A-PD-055-003/PQ-001 Occupation Specialty Specification (AILD) dated 28 Sep 2007

2. A trainee will achieve the qualification AILD when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve the following MITE Course IDs when standards listed in the associated Performance Objectives have been met:

   a. Module 1: Distance Learning - MITE Course ID 123106

   b. Module 2:

      1. Residential - 123107
      2. OJPR - 109269

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
Fleet/NFS(A)	Module 1	123106	AILD
NFS(A)	Mod 2-Residential	123107	AILD
Fleet	Mod 2-OJPR	109269	AILD

CHAPTER 1

TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the submarine NCIOP- AILD. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration and 8 periods in a day.

3. **Module 1:** This Module is DL and students should be allowed some freedom to work at their own pace. Times given below are estimations, however students will be limited to 35 calendar days to complete the module.

Course Duration Calculation	
Allocation of ATT (Estimation – time will vary dependent on individual)	**Periods**
Sum of all EOs/POs	8 DL
Enabling Checks (ECs)	11 DL
Performance Checks (PCs)	0
Estimated Actual Training Time (ATT) Sub-total	**19 DL**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	NA
Training Administrative Time (TAT)	NA
Training Travel Time (TTT)	NA
Physical Fitness (PT)	NA
(NA)Sub-total	**NA**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**NA**

4. **Module 2: Residential**

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	99
Enabling Checks (ECs)	15
Performance Checks (PCs)	29

Actual Training Time (ATT) Sub-total	**18 days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	2
Training Administrative Time (TAT)	2
Training Travel Time (TTT) -	0
Physical Fitness (PT)	8
Sub-total	**2 days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**20days**

Note: PT was reduced due to trainer runs allowing the instructor to allocate PT and additional BAT and TAT when student is not participating in a run or PC.

REQUIRED COURSE STAFFING

5. Identify key personnel required to conduct the training.

6. Module 1:

Position	Rank	Qualifications	Qty	PO / EO
Course Officer	Lt(N)	BMOQ	1	ALL
Senior Instructor	PO2/PO1	BSQ, IS, AIT	1	ALL
Instructor	MS/PO2/PO1	AILS, AILD, AIT,	1	ALL

7. Module 2:

Position	Rank	Qualifications	Qty	PO / EO
Course Officer	Lt(N)	BMOQ	1	ALL
Senior Instructor	PO2/PO1	BSQ, IS, AIT	1	ALL
Instructor	MS/PO2/PO1	AILS, AILD, AIT,	1	ALL
Instructor	Cpl and above	QL5 Image Tech	1	EO 004.01
Trainer Operator (VCSTT)	Any and Civ	VCSTT Operator Course	1	PO 003
Trainer Operator (Nav Trainer)	Any and Civ	Navigation Trainer Operator	1	PO 001
Guest Speaker	PO2/PO1/CPO2	AILS, AILD, NCIOP RQ PO2	1	PO 002

COURSE CAPACITY

8. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

9. Module 1:

a. Maximum – 8;

b. Minimum – Nil; and

c. Optimum – Nil.

10. Module 2:

d. Maximum – 6;

e. Minimum – 2; and

f. Optimum – 4.

PREREQUISITES

11. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

12. Module 1:

g. NCIOP RQ-LS (ALAU);

h. Minimum, Level 2 Security Clearance;

i. Basic Submarine Qualification (BSQ) Module 1 (AILS - 109201); and

j. NAV Yeoman (AGXO).

13. Module 2:

k. Module 1 (MITE 123106);

l. NCIOP RQ-LS (ALAU);

m. Minimum, Level 2 Security Clearance;

n. Basic Submarine Qualification (BSQ) Module 1 (AILS - 109201); and

o. NAV Yeoman (AGXO).

COURSE REPORTING

14. Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or Locally Produced Equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

15. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

16. All attempts should be made to schedule the sub NCIOP course (AILD) in close proximity to the NAV Yeoman course (AGXO). The sequencing of the two courses is not critical, however scheduling both courses consecutively will streamline training in order to achieve effective employment as an NCIOP on board Victoria Class Submarines.

17. This course includes a DLN portion (Module 1). Students are to be loaded on Module 1 on completion of BSQ and units are to permit sufficient time for the student to complete the DLN module in a quiet environment to allow maximum concentration.

18. Prior to Module 2, the candidate will complete the pre-instruction package (PIP). The **PIP will be completed no more than 45 days prior to Module 2** (Residential).

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – Perform the Duties of a Radar Watch Keeper

1. **Performance**: Perform the duties of a radar watch keeper onboard a Victoria class submarine.

2. **Conditions**:

   a. Given:

      (1) Navigation Trainer;
      (2) Operational Submarine;
      (3) Operational Radar Equipment; and
      (4) Access to DLN.

   b. Denied: Nil.

   c. Environment:

      (1) At Sea or Alongside
      (2) All Weather Conditions
      (3) Surfaced or Submerged Victoria Class Submarine; and
      (4) All Lighting Conditions.

3. **Standard**: The NCIOP shall perform the duties of a radar watch keeper Onboard a Victoria class submarine by:

   a. Performing flash-up procedures (K0001) to include;

      (1) Flashing-Up (K0003), Tune Radars, and Displays(C0075) at Sea and Alongside; and
      (2) Calculating Radar Index Error.

   b. Conducting daily user checks (C0005);

   c. Operating radars (D0040/D0045/T0001) to include;

      (1) Responding to a Man Overboard EOP;
      (2) Tracking Contacts;
      (3) Performing Anti-Collision Tracking;
      (4) Ranging Targets; and
      (5) Performing Radar Cut Procedures (C0215).

   d. Reporting all contacts in accordance with control orders (D0020/T0003); and

   e. Performing shut down procedures.

4. **Training Limitations**: Nil.

EO 001.01 - Flash Up and Shut Down the Submarine Radars

LG 001.01.01 - Flash Up and Shut Down the Submarine Radars

Conditions		Sequence
DLN		Beginning of PO 001
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Flash up and shut down the submarine Radars to include: 1) Caps and Lims; 2) Radar Components; 3) Portable Radars; 4) Radar States; 5) Radar Flash up; and 6) Radar Shut Down.		DLN/ Interactive Lectures/ Video

Scenario	Multiple. Integrated in DLN Content.
Activity Steps	1) Student will read through DLN content. 2) Student will answer all questions in the confirmation checks throughout the DLN content. 3) Student will review the material prior to completing the EC at the end of the EO.

Resources	References
1) EO 001.01.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) NCPM 401(B) 2) C-59-875-G00\MB-001 Color Tactical Display 3) DLN Content

Notes to Instructor
In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

EC 001.01- Flash Up and Shut Down the Submarine Radars

Scope	Conditions	Sequence
EO 001.01	DLN	After EO 001.01
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)

Scenario	Integrated in DLN content.
Assessment Steps	1) Multiple choice test, conducted on DLN. 2) The student must achieve a minimum mark of 70 % in order to be successful.

Resources	References
1) EC 001.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) NCPM 401(B) 2) C-59-875-G00\MB-001 Color Tactical Display 3) DLN Content

Notes to Instructor
Performance on equipment will be assessed as part of the OJPR. In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

Limitations	Nil
Impact of Failure	No impact to failure. The student will, on their own, review the content of EO 001.01 and attempt the test until successful. If the student requires amplifying information, they will contact the instructor for further explanations.

EO 001.02 - Track Contacts and MOB Procedure

LG 001.02.01 – Track Contacts and MOB Procedure

Conditions		Sequence
Navigation Trainer S-17		First Residential
Time	**Classification**	**Instructor**
8 x 50 min (2x50 min periods and 1x50 min period per student)	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Operate the submarine Radars to include: 1) Calculate Radar Index 2) Contact Tracking 3) Reporting Procedure 4) Man Overboard		EDI

Scenario	You are the NCIOP on watch during a fog routine exiting harbour. In order to maintain safety of the submarine you need to maintain a watch on radar IAW control orders.
Activity Steps	1. The instructor will give the scenario to provide context to students. 2. The instructor will review the Radar controls. 3. The instructor will demonstrate how to calculate the radar index error. 4. The students will calculate the radar index error with the instructor, individually. 5. The instructor will explain and demonstrate the radar operation, starting by creating a new radar contact and reporting all information to the Plot officer. 6. The instructor will review all associated voice reports. 7. The instructor will then demonstrate the man overboard procedure as a radar watch keeper. 8. The students will individually track a radar contact and report. The students will conduct man overboard procedures with the instructor. 9. The other students will follow along with the material being covered. 10. The instructor will conduct a review with the students. The review shall be an interactive format. The instructor will ask each student as a

	group what the tactical implication of each point covered should be for the submarine and when it would be used. The aim of the review is to ensure the students understand the subject covered and its importance by reinforcing the tactical implications that Radar employment has on the submarine.

Resources	References
1) Navigation radar with submarine specific display	1) NCPM 401(B) 2) VCSOPs 3) C-59-875-G00\MB-001 Color Tactical Display

Notes to Instructor

The students will gather around the display to allow everyone to see the complete procedure. If the course is composed of more than three students, each demonstration will be done twice by the instructor, to permit all students to see and understand the correct procedure.

EC 001.02 – Track Contact and MOB Procedure

Scope	Conditions	Sequence
EO 001.02	Navigation Trainer S-17	After EO 001.03
Time	**Classification**	**Instructor**
6x50 min (1x50min/student)	Unclassified	MS-PO2 NCIOP (subs)

Scenario	Departing NC jetty transiting outbound in Halifax harbour during fog routine. As the NCIOP you will operate radar for anti-collision. The OOW on the bridge has a visibility of approximately 100 yards, all contacts must be tracked and reported.
Assessment Steps	1) The student is to demonstrate their ability and knowledge of the VCS navigation radar. The student is to set up and operate radar in accordance with references, by reacting to events during a simulation. 2) The student is to demonstrate their ability to calculate the radar error before departure. 3) The scenario will be based on H.M.C.S. Windsor heading outbound toward George's Island in reduced visibility. The assessor will act as the Plot Officer for all interactions with the student. 4) The student will maintain tracking on all vessels and make all reports to the Plot Officer. 5) Once the first two sections of EC 001.02.01 have been assessed, the instructor will initiate an MOB, the student must react accordingly. 6) Instructor will assess using the EC checklist and provide feedback to the individual student after each scenario in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) Navigation radar with submarine specific display 2) EC 001.02 check list	1) C-59-875-G00\MB-001 Color Tactical Display 2) VCSOPs 3) NCPM 401 (B)

Notes to Instructor
MOB is a critical on this run. If not properly executed, it is an automatic fail for safety. There is no retest for this EC, it will be tested on the PC. The instructor assessing you will act as the Plot Officer.

Limitations	Nil
Impact of Failure	No impact to failure. Instructor to determine, with the student, if extra training time is required.

EC 001.02 Track Contacts and MOB Procedure

Item	EC 001.02 Track Contacts and MOB Procedure	SAT	UNSAT	Remarks
1. Detection				
Create a new radar contact				
Initial report is clear				
The Information Passed to PLOT is accurate				
2. Maintain Tracking				
Tracking was maintained				
Reported CPA was accurate				
3. Man Overboard				
Inputting MOB symbology in display **(CRITICAL)**				
Reporting bearing and range (X3)				

Score: /7

Trainee must achieve 5/7 SATs on all factors **AND** meet all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.03 - Perform the Radar Cut Procedure

LG 001.03.01 Perform the Radar Cut Procedure

Conditions		Sequence
Navigation Trainer S-17		After EO 001.02
Time	**Classification**	**Instructor**
4 x 50 min (1x50 min periods; and 1x50 min periods per two students)	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Operate the submarine Radars for a radar cut procedure.		EDI

Scenario	You are the NCIOP on watch during a return to PD. In order to maintain safety of the submarine you need to close up a radar watch and perform Radar Cut Procedures.
Activity Steps	1. The instructor will review EO 001.02 with the students individually and collectively, asking relevant questions about the steps covered in EC 001.02. 2. The instructor will demonstrate a radar cut procedure as per NCPM 401B. This will be demonstrated with 4 scenarios: a. Radar cut with a contact within 1500 yards of the submarine b. Radar cut with a contact between 1500 yards and 3000 yards of the submarine c. Radar cut with a contact outside 3000 yards of the submarine; and d. Radar cut with no contact. 3. The instructor will one at the time have each student go through all possible scenarios of radar cut in no particular order. 4. The instructor will ask if there are any questions and cover any points/issues students had trouble with during scenarios.

Resources	References
1) Navigation radar with submarine specific display	1) NCPM 401 B 2) C-59-875-G00\MB-001 Color Tactical Display

Notes to Instructor
The students will gather around the display so all can see the information presented. If the course is composed of more than 3 students each demonstration will be done twice by the instructor to permit all students to see and understand the information taught to them. If the students are having difficulty meeting standard, allow extra time for the students to practice this knowledge. The use of Fred Sherwood may need to be considered if the students require extra training or if the navigation trainer is unavailable.

EC 001.03– Perform a Radar Cut Procedure

Scope	Conditions	Sequence
EO 001.03	Navigation Trainer S-17	After EC 001.02
Time	**Classification**	**Instructor**
6x50 min (1 period per student)	Unclassified	MS-PO2 NCIOP (subs)

Scenario	The submarine is returning to PD and command has requested an NCIOP to operate radar prior to surface IAW NCPM 401 B. You must keep the submarine safe during the Radar cut procedure.
Assessment Steps	1) The student is to demonstrate their ability and knowledge on the Radar cut procedure. The student has to set up the radar display and operate radar in accordance with references by reacting to events in a simulated environment. 2) The student will be allowed 10 minutes to set the radar display IAW references. 3) The assessor will act as Plot Officer for the duration of the run. 4) Once ready, the instructor will simulate a return to PD at which point the student will be required to react by performing a radar cut procedure. 5) The assessment will include four radar cuts to ensure all variations of the procedure are assessed. (contact within 1500 yds, contact between 1500 and 3000 yds, contact outside 3000 yds and no contact) The order in which the radar cuts are conducted will be determined by the instructor. 6) Instructor will provide feedback to the individual student, after each scenario in order to review the drill and reinforce correct operating procedures.

Resources	References
1)EC 001.03 Checklist 2) Navigation radar with submarine specific display	1) C-59-875-G00\MB-001 Color Tactical Display 2) NCPM 401 B

Notes to Instructor
For safety reasons, any radar cut procedure with a contact within 3000 yards is critical, in the event of a failure, provide additional training prior to the PC.

Limitations	Nil

Impact of Failure	No impact to failure. Instructor to determine, with the student, if extra training time is required.

EC 001.03 - Perform a Radar Cut Procedure

Item	EC 001.03 Perform a Radar Cut Procedure	SAT	UNSAT	Remarks
Radar Cut Procedure, Contact Within 3000 Yards (Critical)				
Set-up display for a radar cut procedure to include: • Select the appropriate range (3000yds) • Set correct range on VRM (1500yds)				
Track and report a contact within 1500 yds. IAW NCPM 401B				
Track and report a contact between 1500 to 3000 yds. IAW NCPM 401B				
Radar Cut Procedure, Contact Outside 3000 Yards				
Track and report a contact outside 3000 yds.				
Report no contact is held.				

Score: /5

Trainee must achieve 3/5 SATs on all factors **AND** an SAT on all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 001 – Perform the Duties of a Radar Watch Keeper

Scope	Conditions	Sequence
PO 001	Navigation Trainer S-17	After EC 001.03
Time	**Classification**	**Instructor**
11x50 min (8x 50 min and 3x50 min rewrites)	Unclassified	MS-PO2 NCIOP (subs)

Scenario	The submarine is returning to PD and command has requested an NCIOP to operate radar prior to surface in order to clear any possible threat.
Assessment Steps	1) The student is to demonstrate their ability and knowledge of the Kelvin Hughes navigation radar and the radar cut procedure. The student shall set up their radar display and operate radar in accordance with C-59-875-G00\MB-001 Colour Tactical Display and NCPM 401(B), by reacting to events in a simulated environment. 2) The student will have time to set the radar display as per NCPM 401(B). 3) Once ready the instructor will simulate a return to PD at which point the student will have to react by performing a radar cut procedure. 4) The instructor will act as the Plot Officer for the assessment. 5) As the submarine returns to PD the instructor will assess the radar cut procedure with a contact within 1500 yds of the submarine. 6) The submarine will lower all masts and proceed deep. 7) The submarine will then attempt another return to PD, a contact will be between 1500 yds and 3000 yds, the contact must be past CPA and opening. 8) The student will track contact until safe to surface. Once surfaced, the student will keep monitoring the contact and report to plot. Two other contacts must be added during the run. Students must maintain tracking for anti-collision purposes. 9) Once the instructor has been able to assess all the points from the PC checklist, except the MOB, the instructor must simulate an MOB as the final point in assessment. 10) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures.

Resources	References
1) Navigation radar with submarine specific display 2) PC 001 checklist	1) C-59-875-G00\MB-001 Color Tactical Display 2) VCSOPs 3) NCPM 401 (B)

Notes to Instructor
The radar cut procedure and the MOB are critical to this run. Review the PC on completion of each run with the assessed student. Provide the student with constructive feedback.

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PC 001 Perform the Duties of a Radar Watch Keeper

Item	PC 001 Perform the Duties of a Radar Watch Keeper	SAT	UNSAT	Remarks
Radar Cut Procedure (Critical)				
Set-up display for a radar cut procedures to include: • Select the appropriate range (3000yds) • Set correct range on VRM (1500yds)				
Track and report a contact within 1500 yds of the submarine.				
Track and report a contact between 1500 yds and 3000 yds of the submarine.				
Operate the Submarine Radar as a PLOT Assist.				
Detect and track a minimum of 3 radar contacts to include: • Initial detection and report • Maintain tracking for anti-collision • Report CPA to PLOT				
Conduct a man overboard procedure **(CRITICAL)**				

Score: /8

Trainee must achieve all critical factors to pass this PC.

Positive Feedback	Improvement Feedback

NTDC(A) Approval – 03/02/2021
NPTG HQ Approval - DDMMYYYY
Version No: 2.1

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 – Conduct a Mission Brief

1. **Performance**: Conduct the NCIOP portion of a mission brief.

2. **Conditions**:

   a. Given:

      (1) Operational Submarine; and

      (2) Access to Intelligence Information.

   b. Denied:

      (1) Nil

   c. Environment:

      (1) At Sea or Alongside
      (2) All Weather Conditions
      (3) Surfaced or Submerged; and
      (4) All Lighting Conditions.

3. **Standard**: The NCIOP, as an Electronic Warfare Operator, shall conduct the NCIOP portion of a mission brief onboard a submarine by:

   a. Building Platform Cards(C0015/C0025/C0035/C0130);
   b. Building a Threat Matrix (C0040);
   c. Building Height Cards (C0035);
   d. Building a Threat Binder (T0004);
   e. Preparing the Information using Jane's (C0130);
   f. Briefing Command on Expected Platform (B0020/C0040); and
   g. Briefing Command on Threat Radars (B0020).

4. **Training Limitations**: Nil

EO 002.01 -Build the NCIOP Portion of a Mission Brief

LG 002.01.01 - Building the NCIOP Portion of a Mission Brief

Conditions		Sequence
DLN		After EO 001.01
Time	**Classification**	**Instructor**
1x50 min	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Building an NCIOP portion of a mission to include: 1) Platform Cards; 2) Height Cards; 3) Threat Matrix; and 4) Threat Binder. Using open sources, the student must be able to collect platform information to support the submarine operations.		DLN

Scenario	Multiple. Integrated in DLN content
Activity Steps	1) Student will read through DLN content. 2) Student will answer all questions in the confirmation checks throughout the DLN content. 3) Student will review the material prior to completing the EC at the end of the EO.

Resources	References
1) Jane's Online 2) Weyers Flotten Taschenbuch 3) DLN Content 4) Computer (For DLN access)	1)NCPM 401 B

Notes to Instructor
There will be an EC for this EO. The student shall contact the instructor to receive their threat list and assessment. The student shall have access to a blank form in order to complete their EC. In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

EC 002.01- Build the NCIOP Portion of a Mission Brief

Scope	Conditions	Sequence
LG 002.01.01	DLN	After LG 002.01.01
Time	**Classification**	**Instructor**
8x50 min	Unclassified	MS-PO2 NCIOP (subs)

Scenario	You are preparing for an upcoming mission aboard the submarine, you must provide your combat officer with a mission threat brief.
Assessment Steps	1) The student will be provided with a scenario and blank form. 2) Using the blank form provided to them, the student shall fill in the information on the threats assigned to them. It must include the 2 surface cards and height cards, 2 aircraft cards, 1 submarine card, 1 recognition binder and 1 threat matrix. The threat matrix and recognition binder must include all the platforms included in the scenario plus 10 platforms that could be in the area of the scenario. The student must make their own list of 10 platforms based on the information contained in the scenario. 3) Once the student has completed the project, they will send it to the course instructor via DWAN email for corrections. 4) The instructor will assess in accordance with the marking sheet EC 002.01.02. A copy of the marking sheet will be sent by the instructor to the student with feedback 5) If the student passes, the power point will be saved by the instructor for use during module 2. If the student fails to achieve the minimum standard of 40/56, the student will make corrections and re-submit to the instructor. 6) Upon re-submission, the instructor will assess in accordance with the marking sheet EC 002.01.02. A copy of the marking sheet will be sent by the instructor to the student with feedback. If the student passed the power point, it will be saved by the instructor for use during module 2. If the student fails to achieve the minimum standard of 40/56, the instructor will contact the student and their supervisor in order to provide extra training. Another scenario may be used for a new project if more training is required.

Resources	References
1) Scenario Pack 2) Blank threat pack 3) Computer	1) NCPM 401 B

Notes to Instructor
This will be done as part of the DLN portion of this course and sent via email for assessment. The material must be saved for future use in the residential phase of this course. The scenario shall be selected from the pack of scenarios with the platform list.

Limitations	Nil
Impact of Failure	No impact to failure. Instructor to liaise with the student to determine, if extra training is required. The student shall be allowed a minimum of two attempts to achieve the passing mark, after the first attempt the instructor shall provide feedback to enable the students to know where the marks have been missed. If unsuccessful on the second attempt the instructor will contact the student and their supervisor to explore what extra training is needed.

EC 002.01 Build the NCIOP Portion of a Mission Brief

Item	EC 002.01 Build the NCIOP Portion of a Mission Brief	SAT	UNSAT	Remarks
Surface Platform #1				
Top row displays proper labeling				
Picture is accurate				
All information on the card is filled in with accurate information to include:				
• Speed, Draught, Sec Ship, Safe Depth and GDR				
• Hulls in class				
• Helo				
• Radars				
• Weapon				
Height Card #1				
Labels are clear				
Virtual height is in increments of 40 virtual				
All information is accurate to include:				
• Max Speed				
• Draught				
• Second Ship				
• Go Deep Range				
Ranging heights are accurate including: • Mast Head • Side Lights • Stern Lights • Height between Lights				
Surface Platform #2				
Top row displays proper labeling				
Picture is accurate				
All information on the card is filled in with accurate information to include:				
• Speed, draught, second ship, safe depth and GDR				
• Hulls in class				
• Helo				
• Radars				
• Weapon				
Height Card #2				

Labels are clear			
Virtual height are in increments of 40 virtual			
All information is accurate to include:			
• Max speed			
• Draught			
• Second ship			
• Go deep range			
Aircraft Platform #1			
Top row displays proper labeling			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Aircraft Platform #2			
Top row displays proper labeling			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Submarine Platform			
Top row displays proper labeling			
Surface speed is accurate			
Dived speed is accurate			
Picture is accurate			
Displacement surfaced and dived is accurate			
Propulsion is accurate			
ESM system is accurate			
Radars are accurate			
Threat Matrix			
Includes all 10 possible encounters and all platforms from the brief			
Emitters are easily identified with the associated platform			

Matrix is broken down into section easy to read			
All radars are included			
Recognition Binder			
Easy to follow			
Key components are highlighted			
All platforms are included			
All platforms are properly labeled			
All pictures display proper markings (flag, hull number)			

Score: / 56

The trainee must achieve 40/56 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 – Brief the NCIOP Portion of a Mission Brief

LG 002.02.01– Briefing the NCIOP Portion of a Mission Brief

Conditions		**Sequence**
Classroom		After PC 001
Time	**Classification**	**Instructor**
4X50 min periods	Unclassified	PO2-CPO2 NCIOP (subs) / Guest Speaker
Learning Objectives		**Method**
Briefing command to include: 1) Electronic warfare; 2) Submarine tactics against radar; 3) Understanding the radar environment and threats and 4) Briefing techniques.		Briefing

Scenario	Nil.
Activity Steps	1) A guest speaker will talk about submarine related Electronic Warfare and briefing techniques relating to submarine briefs. The speaker will explain to the students what is expected of them when briefing command. This is an opportunity for the students to be introduced to submarine tactics and have a better understanding of what a radar environment is and what to expect from threat radars. The speaker will be present for 2 periods. 2) On completion of the brief from the guest speaker, the student will be given one period to review their brief from LG 002.01.02 prior to the EC. 3) As part of the brief review the student shall adapt their brief to reflect the knowledge taught to them by the guest speaker. 4) For the last period of LG 002.01.03 the instructor will give a new surface platform and an aircraft platform to be added to the brief. The student will be denied access to DWAN connection to simulate a watch at sea. The student will produce a surface card, a height card and an aircraft card. The student will also brief the change to the threat matrix and recognition binder with regards to the new platform. The only material available to them will be Jane's book, Weyers and MEWD. The students may work in groups of 2.

Resources	**References**
1) Guest Speaker (EW1 Qualified) 2) EC 002.01.01 briefs	1) NCPM 401 B

Notes to Instructor

The instructor will insure they have all the EC 002.01.01 briefs prior to commencement of this EO. The briefs will be used to present for the EC.

EC 002.02 – Brief the NCIOP Portion of a Mission Brief

Scope	Conditions	Sequence
EO 002.01	Classroom	After LG 002.01.02
Time	**Classification**	**Instructor**
3 x 50 min period (1 per 2 students)	Unclassified	MS-PO2 NCIOP (subs)

Scenario	You are preparing for an upcoming mission aboard the submarine, you have been tasked by your combat officer to deliver a mission threat brief.
Assessment Steps	1) The student will brief a command team. Each student will present their own brief. They will have access to the brief built during EO 002.01. The aim of this EC is to test the knowledge on radar theory and EW against counter detection. 2) The instructor will act as the OPSO and will introduce the scenario given to the students and to members attending the brief. 3) The student will brief all assigned vessels, submarine, and aircraft given to them by the instructor. The student will compile information and brief command on the platform assigned to them. The brief shall include radars and major weapons carried that could harm a submarine at surface with the exception of torpedoes. The student will be tested on the material and not on the delivery of the brief. 4) The student will include recommendations to avoid counter detection. 5) Members attending and the instructor will question any information on which they need more clarifications and provide feedback to the individual student. After each scenario is completed the instructor will review the marking guide and provide more feedback if required in order to reinforce good briefing habits.

Resources	References
1) One SMORO Qualified Officer 2) One CHOPS Qualified PO1 3) One EW1 Qualified PO2 4) Access to DWAN 5) Projector or TV Screen 6) Briefing Room	1) NCPM 401 B

Notes to Instructor
The recommendation to command must be pertinent, if in doubt you may question the student to get more clarity on the recommendation. The location of this EC must be suitable for a

command brief to include a working projector or TV screen and enough seating for all attendees. The ECs are to be done one at a time. Other students shall be in a different room while awaiting their EC. The instructor is to conduct a one on one de-briefing with each student on completion of their EC.

Limitations	Nil
Impact of Failure	No impact to failure. Instructor to determine, with the student, if extra training time and re-briefing is required.

EC 002.02 Brief the NCIOP Portion of a Mission Brief

Item	EC 002.02 Brief the NCIOP Portion of a Mission Brief	SAT	UNSAT	Remarks
1. Threat Information				
Accuracy of the information briefed				
Referenced used				
Clarity of the information				
2. Briefing Command				
The student was prepared				
The student displayed comprehension of the EW environment being briefed				
3. Recommendation against Counter Detection				
Includes one recommendation against detection per platform				
Recommendation is pertinent				

Score: **/ 7**

The Trainee Must Achieve 5/7 SATs to Pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002 – Conduct a Mission Brief

Scope	Conditions	Sequence
PO 002	At home unit	OJPR
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)

Scenario	You are preparing for an upcoming mission aboard the submarine, you have been tasked by your combat officer to deliver a mission threat brief.
Assessment Steps	1) The student will brief a command team. This brief shall be conducted in front of minimum CHOPS, EW and OPSO or XO. The student must build their brief in accordance with their unit's briefing cards. The preparation of the brief must include a threat matrix and a recognition binder. The matrix and binder must be available to the assessing staff at the beginning of the brief. 2) A member of the assessing staff will introduce the scenario given to the students to members attending the brief. 3) The student will brief all assigned vessels and aircraft given to them by their CHOPS. The student will source the information and brief command on the platform assigned to them. The brief shall include radars and major weapons carried that could harm a submarine at surface, with the exception of torpedoes. The student will be tested on the material and the delivery of the brief. The student will include tactics expected from the targets and will include recommendations to avoid counter detection. 4) Members attending will question any information on which they need more clarification and provide feedback to the student. On completion of the brief, EW1 will review the marking guide and provide more feedback as required in order to reinforce good briefing habits.

Resources	References
1) OPSO or XO 2) CHOPS 3) EW1 4) OJPR 5)PC 002 Checklist	1) NCPM 401 B

Notes to Instructor
This is to be done as part of the OJPR.

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PC 002 Conduct a Mission Brief

Item	PC 002 Conduct a Mission Brief	SAT	UNSAT	Remarks
Surface Platform #1				
Labeled correctly				
Picture is accurate				
All information on the card is filled in with accurate information to include: • Speed, draught, sec ship, safe depth and GDR • Hull in class • Helo • Radars • WPN				
Height Card #1				
Labels are clear				
Virtual height are in increments of 40 virtual				
All information is accurate to include: • Max Speed • Draught • Second Ship • Go Deep Range				
Ranging heights are accurate including: • Mast Head • Side Lights • Stern Lights • Height between Lights				
Surface Platform #2				
Labeled correctly				
Picture is accurate				
All Information on the card is filled in with accurate information to include: • Speed, draught, second ship, safe depth and GDR • Hull in class • Helo • Radars • WPN				

Height Card #2			
Labels are clear			
Virtual height are in increase of 40 virtual			
All information is accurate to include: • Max Speed • Draught • Second Ship • Go Deep Range			
Ranging heights are accurate including: • Mast Head • Side Lights • Stern Lights • Height between Lights			
Aircraft Platform #1			
Top row is labeled correctly			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Aircraft Platform #2			
Top row is labeled correctly			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Briefing Command			
The trainee was prepared			
The student displayed comprehension of the EW environment being briefed			
Recommendation against Counter Detection			
Includes one recommendation against detection per platform			
Recommendation is pertinent			
Threat Matrix			
Includes all 10 possible encounters and all platforms from the brief			

Emitters are easily identified with the associated platform			
Matrix is broken down into section easy to read			
All radars are included			
Recognition Binder			
Easy to follow			
Key components are highlighted			
All platforms are included			
All platforms are properly labelled			
All picture display proper marking (flag, hull number)			

Score: / 39

The trainee must achieve 28/39 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
EW1	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 – Perform the Duty of an Electronic Warfare Operator

1. **Performance**: Perform the duties of an electronic warfare operator

2. Conditions:

a. Given:

(1) Operational Submarine; and
(2) Operational ESM Suite.

b. Denied:

(1) Nil.

c. Environment:

(1) At Sea or Alongside
(2) All Weather Conditions
(3) Surfaced or Submerged Victoria Class Submarine; and
(4) All Lighting Conditions.

3. **Standard**: The NCIOP shall perform the duties of an electronic warfare operator by:

a. Performing a proper flash up the ESM suite (K0002/C0075) to include:

(1) Flashing Up the ESM Suite (K0003);
(2) Loading the Proper Libraries (B0045/T0009);
(3) Rebooting the ESM Suite; and
(4) Shutting Down the ESM Suite

b. Conducting confidence checks (C0005)

c. Operating the electronic support measures (ESM), equipment (C0210/T0002) to include:

(1) Detecting a Signal (D0020/ K0008/T0010);
(2) Analyzing a Signal (T0011);
(3) Collecting a Signal(C0080/D0045);
(4) Conducting a BINT (B0040); and
(5) Operating the EW Equipment in Simulation Mode.

d. Conducting ESM reporting procedures (C0195/T0007/T0008) to include:

(1) Applying Control Orders;

(2) Conducting Initial and Amplifying Reports;
(3) Conducting Warner Clearance and Warner Look Procedures;
(4) Conducting Dangerous Racket Reports;
(5) Applying Danger Level in Different Mast States; and
(6) Maintaining the EW Log (C0065/T0005).

4. Training Limitations: N/A

EO 003.01 - Flash Up the Electronic Warfare Equipment

LG 003.01.01 – Flash Up the Electronic Warfare Equipment

Conditions		Sequence
DLN		After LG 002.01.01
Time	**Classification**	**Instructor**
1 x 50 min	Unclassified	MS-PO2 NCIOP (subs)

Learning Objectives	Method
Describe the electronic warfare equipment to include: 1) Caps & Lims; 2) EW Components; 3) Flash Up; 4) Reboot; 5) Confidence Checks; and 6) Shut Down.	DLN

Scenario	Multiple, integrated in DLN content
Activity Steps	1) Student will read through DLN content. 2) Student will answer all questions in the confirmation checks throughout the DLN content. 3) Student will review the material prior to completing the EC at the end of the EO.

Resources	References
1) EO 003.01.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

EC 003.01 – Flash Up the Electronic Warfare Equipment

Scope	Conditions	Sequence
EO 003.01	DLN	After EO 003.01
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)

Scenario	Integrated in DLN content.
Assessment Steps	Multiple choice test conducted on DLN. The student must achieve a minimum mark of 70 % in order to be successful.

Resources	References
1) EC 003.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

Limitations	Nil
Impact of Failure	No impact to failure. The student will on their own review the content of EO 003.01 and do the EC until successful. If the student requires amplifying information, they will contact the instructor for further explanations.

EO 003.02 – Conduct ESM Reporting Procedures

LG 003.02.01 – ESM Reporting Procedures

Conditions		Sequence
DLN		After EC 003.01
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Conduct ESM reporting procedure to include: 1) Applying Control Orders; 2) Conduct Initial and Amplifying Reports; 3) Conduct Warner Look and Warner Clearance Procedures; 4) Dangerous Racket Procedure; 5) Apply Danger Levels in Multiple Mast States; and 6) Maintain the EW Log.		DLN

Scenario	Multiple, integrated in DLN content
Activity Steps	1) Student will read through DLN content. 2) Student will answer all questions in the confirmation checks throughout the DLN content. 3) Student will review the material prior to completing the EC at the end of the EO.

Resources	References
1) EO 003.02.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
This portion will be evaluated during the developmental runs, final assessment during PC. In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

EO 003.03 – Operate ESM Equipment

LG 003.03.01 – Radar Theory

Conditions		Sequence
DLN		After EO 003.02
Time	**Classification**	**Instructor**
2x 50 min	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Describe the RADAR fundamentals to include: 1) Components; 2) Range; 3) Scan Types; 4) Characteristics; and 5) Parameter Effects on Performance;		DLN

Scenario	Multiple integrated in DLN content
Activity Steps	1) Students will read through DLN content. 2) Students will answer all questions in the confirmation checks throughout the DLN content. 3) Students will review the material prior to completing the EC at the end of the EO.

Resources	References
1) EO 003.03.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

EC 003.03 – Radar Theory

Scope	Conditions	Sequence
EO 003.03.01	DLN	After LG 003.03.01
Time	**Classification**	**Instructor**
1x 50 min	Unclassified	MS-PO2 NCIOP (subs)

Scenario	Integrated in DLN Content.
Assessment Steps	Multiple choice test conducted on DLN. The student must achieve a minimum mark of 70 % in order to be successful.

Resources	References
1) EC 003.03 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
This portion is critical for the completion of this course, in order to complete the DLN portion the student must pass this EC. In the event the student needs further information, the instructor will use all available resources to provide the information to the students.

Limitations	Nil.
Impact of Failure	No impact to failure. The student will on their own review the content of EO 003.03.01 and do the EC once again. If the student requires amplifying information, they will contact the instructor for further explanations.

LG 003.03.02 – Initial Setting of the ESM suite

Conditions		Sequence
Fred Sherwood Trainer (S-17)		After EC 002.01
Time	**Classification**	**Instructor**
6 x 50 min period	Secret	MS-PO1 NCIOP (subs)
Learning Objectives		**Method**
Operate the EW equipment to include: 1) Set-Up ESM Suite for Mission Specific; 2) Conduct Confidence Checks;		EDI

Scenario	The submarine has piped standby to return to PD. You must set up the ESM suite for EW operations.
Activity Steps	1. The instructor will start this lesson with a quiz as a review of material seen on DLN and to test knowledge of the students with regards to EW. This quiz will be reviewed as a class and is not marked. 2. The instructor will cover the operation and set-up pages used in the ESM suite. 3. The instructor will explain and demonstrate, with visual material (video), the proper ESM suite set-up to include: - Purge Time - Purge Level - Tactical Level - PRF-PRI - File Logging - Gated Audio -Library set up to include: - Choosing a Library - Saving the Preset - Applying Library -Confidence checks to include: - Summary BIT/CAL - Antenna BIT

	- RPU BIT - Antenna CAL - End-to-End Test 4. The students will, one at the time, sit at the ESM suite to go through the operation pages and proper set-up as demonstrated. The remainder of the students shall have access to the live stream of ESM suite in the Fred Sherwood Theater. 5. The instructor will review the content covered in the lesson. The instructor will ensure the student understands the material cover by asking each student what the implication would be if each part of the set-up is not completed correctly.

Resources	**References**
1) EW Suite Trainer 2) Fred Sherwood Theater 3) Fred Sherwood Trainer 4) EW Quiz	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
It is important that they can perform these tasks independently, this is the foundation of EW operation and will provide a solid base to build on as an EW operator.

LG 003.03.03 – Basic EW Operation

Conditions		Sequence
Fred Sherwood Trainer (S-17)		After LG 003.03.02
Time	**Classification**	**Instructor**
6 X 50 min periods	Secret	MS-PO1 NCIOP (subs)
Learning Objectives		**Method**
Operate the EW equipment to include: 1) Recognizing the Mast Exposure; 2) Detection of an EW Signal; 3) Analysis of an EW Signature; and 4) Brief Command on Threat Radars.		EDI

Scenario	Upon return to PD the search periscope is raised, you have to update command with accurate information with regard to the EW environment.
Activity Steps	1. The instructor will gather all students in the Fred Sherwood Theater. Using the screens to display the EW suite screen the instructor will educate the students on the initial breakdown of an EW racket. The instructor will break down all column of the activity page to include, NUM, T#, TL, CAT, BRG, AMP, RF, PRF, PW, SCAN, TFS,TSLS, LIB ID, RADAR and PLAT 2. The instructor will quiz the students with the information passed to them on the activity page to ensure the students understand what was covered and allow them to ask questions relating to the information covered. 3. The instructor will explain all available tools in the activity page to the students to include, Scan toggle, Create Lib, Delete Entry, Delete All, Delete Old, Manual TARGET Toggle, Exclude TARGET Toggle, Order, Log emitter, Gate Toggle and Histogram Emitter 4. The instructor will review all the information covered with a quiz to confirm the students understand the information covered in LG 003.03.03. The instructor will provide clarification as needed. 5. The instructor will explain the process when receiving a new Racket, following the procedure as described in NCPM 401B. As the instructor explains the procedure they will review the voice procedure previously covered during the DLN portion.

	6. The instructor will demonstrate a warner clearance and a warner look with the information displayed on the screen in the theater. The instructor will perform the procedures with Racket dangerous and without racket dangerous. 7. Using different Rackets the instructor will get each student to perform at least one of the procedures until all procedures have been demonstrated by the students a minimum of two times per procedure. 8. The instructor will demonstrate setting up a watch warner and a check clear procedure. The check clear will be done for a clear and a not-clear to change mast state. 9. Using different rackets, the instructor will get each student to perform at least one of each procedure as a group until all procedures have been demonstrated a minimum of two times per procedure. 10. The instructor will simulate multiple scenarios in multiple mast exposure and involve all students in performing all the procedures covered in LG 003.03.03 as a group. The students shall use their acquired knowledge in radar theory and system operation to provide threat assessment and recommendation to avoid counter-detection.

Resources	**References**
1) EW Suite Trainer 2) Fred Sherwood Theater 3) Fred Sherwood Trainer	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
Visual material for an interactive course may be a white board or a video feed from the Fred Sherwood Trainer. If the trainer is not used by the other courses, the best delivery methods will be to raise the periscope and have a live feed of the ESM in the Fred Sherwood Theater inputting rackets in the system and have a class participation. Ensure all students participate and understand what is being demonstrated.

LG 003.03.04 – EW Signals Collection

Conditions		Sequence
Fred Sherwood Trainer (S-17)		After LG 003.03.03
Time	**Classification**	**Instructor**
7x 50 min period	Secret	MS-PO1 NCIOP (subs)
Learning Objectives		**Method**
Operate the EW equipment to include: 1) Building an Operator Library; 2) Transfer Signal to SCS for Collection; and 3) Maintain EW Log.		EDI

Scenario	Search periscope has identified a contact of interest, the CO wants to conduct a BINT on the contact. You will need to complete a radar collect on the vessel's main emitter.
Activity Steps	1. For ease of instruction the instructor will use the screen in the Fred Sherwood Theater. The instructor must use a visual aid to demonstrate how to build an operator library. 2. Once the information has been covered the instructor will let the students practice the steps previously showed. This will be done one on one with the instructor on the ESM suite simulator. 3. After all the students have covered the operator library, the instructor will cover the SCS portion and conducting an ESM collect. The use of visual aid must be used. 4. On completion of all the SCS and ESM collect demonstrations, the students will practice these steps one on one with the instructor on the ESM suite simulator. 5. When all students have been through all the practical steps and are comfortable with the material covered, the instructor will go over the EW log with the entire course. There is no need to practice at this time as NCIOPs have exposure to a log prior to coming down to the submarine. 6. The instructor will conclude with a review of the material by quizzing the course as a whole.

Resources	References
1) EW Suite Trainer 2) Fred Sherwood Theater 3) Fred Sherwood Trainer	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor
While covering the information with the students, one on one. The screen in the theater shall be selected for a live feed of the ESM suite to allow the other students to review the information on their own.

LG 003.03.05 – Developmental Runs

Conditions		Sequence
Fred Sherwood Trainer (S-17)		After LG 003.03.04
Time	**Classification**	**Instructor**
48x50 min periods	Secret	MS-PO1 NCIOP (subs)
Learning Objectives		**Method**
Combine all EW related topics covered in EO 003.03		EDI

Scenario	As a junior NCIOP onboard the submarine, you will need to perform the duty of Warner during a regular watch at sea responding to the EW environment.
Activity Steps	1) The students will have at least 4 runs each increasing the complexity of the run as they progress. Each run is two 50 min periods. 2) The instructor will monitor them and provide feedback. The instructor shall quiz them on the "racket" being tracked to test their understanding of the EW environment. The instructor will follow the marking sheet in order to provide valuable and pertinent feedback on their run and areas of improvement. 3) The EW log must be filled in as the student is conducting its run 4) At this point the instructor will assess the student to provide feedback on the run. The control room will consist of a minimum one officer of the watch and one CDC operator. In those runs the student will provide recommendations to command based on the EW environment (how to remain undetected at PD, recommendation on mast exposure and analysis of threat emitters). Each student shall have a minimum of 4 developmental runs. 5) Instructor will provide feedback to the individual student after each scenario in order to review the tasks, and reinforce correct operating procedures

Resources	References
1) EW Suite Trainer 2) Fred Sherwood trainer 3) CDC Operator Qualified 4) Scenario Pack 5) PC 003 Checklist	1) CS-6700-14 AN/ULR 502 Sea search Operators Manual 2)NCPM 401 B

Notes to Instructor

The aim of this course is to challenge the students in order to produce more knowledgeable operators. If they are comfortable during the runs, the level of complexity should be increased. It is acceptable to deviate from the scenario provided it is to the benefit of the student. The run should not be easier as the aim of the runs is to prepare them for a watch at sea. There will not be an EC for EO 003.03. All developmental runs must be marked as an EC using the PC checklist and used as formative evaluations. **A student can be unsuccessful in all 4 runs and still attempt the PC**, however instructor and student should work on areas of weakness. If time is available, extra developmental runs can be conducted.

During LG 003.03.05 the instructor shall allow for 3 periods a week PT when a student is not on a development run. It is at the instructor's discretion to allow BAT, TAT, extra study time, etc. during other times when a student is not in the trainer.

PC 003– Perform the Duty of an Electronic Warfare Operator

Scope	Conditions	Sequence
P0 003	Fred Sherwood Trainer / Theater	After LG 003.03.05
Time	**Classification**	**Instructor**
18x50 min (12x 50 min and 6x50 min rewrite)	Secret	MS-PO2 NCIOP (subs)

Scenario	You are sailing as a Junior EW operator, the EW1 is on the helm. The watch is proceeding to PD in order to copy the broadcast.
Assessment Steps	1) The student will be given the 10 minutes to complete the set-up of the ESM suite including the library set-up, mission set-up, and confidence checks. Once setup is complete, the submarine will proceed back to PD. 2) Upon return to PD a warner clearance will be conducted. When warner clearance is completed and the environment is clear to set watch warner, the EW operator must detect all EW rackets, report all EW rackets and build library entries for all new EW rackets. 3) In order to copy the broadcast, warner will conduct a check clear procedure. 4) When a racket becomes dangerous, the EW operator will carry the appropriate actions. 5) The OOW will lower all masts, the EW operator will be asked to brief command on the radar threat. Command may ask questions to clarify the environment and radar threat. 6) With a radar threat the submarine will close up for a BINT on the threat radar. Going back up conducting a warner look, the EW operator will guide command in on the target for a BINT (including picture and a collect). 7) The operator must maintain a clear and accurate EW tote during the whole run. 8) Instructor will provide feedback to the individual student, after each scenario, in order to review the tasks and reinforce correct operating procedures

Resources	References
1) EW Suite Trainer 2) One Trained Submarine 2OOW 3) One Trained CDC Operator	1) NCPM 401B

4) Scenario Pack	

Notes to Instructor
The level of the runs should be the same for all students and not adjusted based on previous runs. The runs should be challenging enough for all students to have them comfortable to an OTT2 level run. There shall be no prompting from the assessor, but information can be requested from CDC or the OOW. When marking the EW log with accuracy, the students are allowed up to 3 mistakes to be logged, if more than 3 it is UNSAT. On completion of the run the instructor will review the marks with the student and provide them with constructive feedback. PT and admin will be managed by the instructor during this portion of the course as only one student is assessed at a time. Similar to LG 003.03.05.

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PC 003 Perform the Duties of an Electronic Warfare Operator

Item	PC 003 Perform the Duties of an Electronic Warfare Operator	SAT	UNSAT	Remarks
1. Set Up				
Mission set-up				
Library set-up				
Confidence checks completed				
2. EW Clearance				
Conduct a warner clearance **(CRITICAL)**				
Conduct a warner look				
3. Operate ESM Equipment				
Detect EW rackets				
Reports EW racket				
Build a library entry				
Conduct a check clear procedure				
Conduct a dangerous rackets procedure to include: ● Detection ● Reports ● Amplification				
Collect an EW rackets				
4. Brief Command				
Brief command on radar threat				
Provide command with amplifying information on radar of interest				
5. EW Log				
Maintain the EW log with accuracy (no more than 3 errors)				
6. Conduct ESM Reporting Procedure				
Apply control orders				
Conduct initial reports				
Conduct amplifying reports				
Apply danger level in multiple mast exposure				

Score: / 18

Trainees must achieve 11/18 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 004 – Perform Duties of an NCIOP Onboard Submarine

Performance: Perform the duties of an NCIOP onboard submarine

5. Conditions:

a. Given:

(1) Operational Submarine;

(2) Photographic Equipment; and

(3) Access to Intelligence Information.

b. Denied:

(1) Nil.

c. Environment:

(1) At sea or Alongside
(2) All Weather Conditions
(3) Surfaced or Submerged Victoria Class Submarine; and
(4) All Lighting Conditions.

6. **Standard**: The NCIOP shall perform the duties of an NCIOP onboard submarine by :

a. Performing the duties of a submarine photographer (C0135/C0200T0006) by:

(1) Ensuring Proper Camera Setting (K0005/K0006/T0013/T0014);
(2) Producing Sharp Photographic Images (K0004) in Various Conditions
(3) Conducting Periscope Imaging (T0015); and
(4) Cataloging Periscope Imaging (C0205/K0007/T0016/T0017/T0018);

b. Conducting pre-sailing duties to include:

(1) Ensuring Proper Set-Up of HQ1, HQ2 and HQ3;
(2) Ensuring Proper Maintenance of the Audio Recording Equipment
(3) Conducting Bridge Pre-Sailing Inspection; and
(4) Conducting the Pre-Sailing Checklist;
(5) Setting-Up the Submarine for Departure to Include:
(a) EW Set-Up
(b) Control Room Set-Up
(c) Bridge Set-Up

7. Training Limitations: Nil

EO 004.01 – Perform the Duties of a Submarine Photographer

LG 004.01.01 – Perform the Duties of a Submarine Photographer

Conditions		**Sequence**
Formation Imaging		After PC 003
Time	**Classification**	**Instructor**
16 x 50 min periods	Unclassified	Qualified Imagery Tech
Learning Objectives		**Method**
Perform the duties of a submarine photographer to include: 1) Proper camera settings; 2) Produce sharp photographic images in various conditions; and 3) Conduct periscope imaging.		EDI

Scenario	Prior to diving stations, the OOW requires that the camera be set on the periscope and tested. You will set up the camera for future employment at PD and ensure the settings are optimal for the operation. You will be required to produce a test picture.
Activity Steps	1) Instructor to determine pre-existing skills of students by asking for examples of previous photography experience. 2) Instructor will explain the primary principles of photography and the relationships between them, to include: • Aperture; • Shutter speed; and • ISO. 3) The instructor will provide an in-depth description of the various functions available on the cameras and how they affect the photo taken. 4) The instructor will brief the students on digital asset management in accordance with the CAF Unclassified Imagery Instructions (Example Only) and NDSODs. 5) The instructor will demonstrate techniques when taking photographs for: • military functions, promotions and presentations; • technical engineering photography (defect documentation and technical investigations); and • evidence gathering.

	6) The instructor will then demonstrate different methods of taking photographs for the above events. These demonstrations should include: • methods pertaining to all lighting conditions; • the effect of changing shutter speed, ISO and aperture; and • the effects of changing the manual settings on the camera. 7) The student will then take multiple photographs with varied settings (manual setup) in order to better understand the effects on the photograph. Instructor to monitor throughout and provide feedback to the students. 8) The instructor will then ask each student what the most difficult part of the lesson was and how they managed to overcome that difficulty. The students will collectively share those difficulties and solutions, learning from each student's experiences.

Resources	**References**
1) Cameras (similar to the one fitted on Victoria class submarines)	1)NCPM 401B 2) CAF Unclassified Imagery Instructions

Notes to Instructor
The instructor shall use the same model of camera as the submarine uses, if not available a model with similar features as the model carried on Victoria Class Submarines. The PC for this is consolidated within the OJPR.

EO 004.02 – Perform the Pre-Sailing Duties of an NCIOP

LG 004.02.01 – Perform the Pre-Sailing Duties of an NCIOP

Conditions		**Sequence**
DLN		After EC 003.03
Time	**Classification**	**Instructor**
2x 50 min	Unclassified	MS-PO2 NCIOP (subs)
Learning Objectives		**Method**
Conducting the NCIOP pre-sailing duties to include: 1) Proper set-up of the HQ's; 2) Proper maintenance of the audio recording equipment; 3) Bridge pre-sailing inspection; 4) Pre-Sailing checklist; and 5) Setting-up the submarine for departure to include: • EW Set-Up • Control Room Set-Up • Bridge Set-Up		DLN

Scenario	Multiple. Integrated in DLN content
Activity Steps	1) Students will read through DLN content. 2) Students will answer all questions in the confirmation checks throughout the DLN content. 3) Students will review the material prior to completing the EC at the end of the EO.

Resources	**References**
1) EO 001.01.01 DLN Content 2) Headphones or Speakers 3) Computer (For DLN access)	1) NCPM 104 B 2)VCSOP

Notes to Instructor
In the event the student needs further information, the instructor will use all available resources to provide the information to the students. Further training and assessment will be completed on the OJPR.

Task List

Task #	Task Statement	Trg / No Trg	PO
A0005	Perform communication checks	No Trg	N/A
A0010	Configure communications terminal equipment	No Trg	N/A
A0020	Operate C4I inter-ship circuits	No Trg	N/A
A0025	Operate CRYPTO equipment	No Trg	N/A
B0020	Advise on C4ISR	Trg	003
B0040	Manage information transfer	Trg	003
B0045	Manipulate databases	Trg	003
C0005	Perform pre-mission checks on equipment	Trg	001/003
C0015	Conduct intelligence gathering	Trg	002
C0025	Contribute to intelligence-gathering operations	Trg	002
C0035	Display intelligence information	Trg	002
C0040	Disseminate intelligence information	Trg	002
C0045	Operate combat system	No Trg	N/A
C0065	Maintain operational logs	Trg	003/004
C0075	Configure shipboard display	Trg	001/003
C0080	Operate data collection equipment	Trg	003
C0120	Calculate radar refraction prediction ranges	No Trg	N/A
C0125	Coordinate ship’s/units request for information	No Trg	N/A
C0130	Identify CAPS and LIMs of platforms	Trg	002
C0135	Operate imaging equipment	Trg	004
C0150	Adhere to emission control restrictions	No Trg	N/A
C0160	Assist to the preparation of CD&E/OT & E/OPTEST test	No Trg	N/A
C0195	Conduct ESM reporting procedures	Trg	003
C0200	Perform the duties of the submarine’s photographer	Trg	004
C0205	Process digital imagery	Trg	004
C0210	Operate EW equipment	Trg	003
C0215	Conduct radar cut procedure	Trg	001
D0020	Contribute to ASUW tactical picture	Trg	001/003
D0040	Operate radars	Trg	001
D0045	Operate C4I system	Trg	001/003
D0050	Operate multi TDL	No Trg	N/A
D0055	Manage multi tadil network	No Trg	N/A

D0065	Operate IFF	No Trg	N/A
D0070	Maintain recognized maritime picture	No Trg	N/A
D0115	Operate ECDIS	No Trg	N/A
K0001	Characteristics and capabilities of submarine radars and display outfits	Trg	001
K0002	Characteristics and capabilities of submarine ESM outfits	Trg	003
K0003	Flash-up procedures	Trg	001/003
K0004	Digital photography	Trg	004
K0005	Photographic equipment	Trg	004
K0006	Photography rules of composition	Trg	004
K0007	Imaging software	Trg	004
K0008	Interpret EW data	Trg	004
K0009	Identification friend or foe	No Trg	N/A
K0010	Capabilities and limitations of masts and periscopes	No Trg	N/A
K0011	Submarine man aloft procedures	No Trg	N/A
T0001	Operate submarine radars and displays	Trg	001
T0002	Operate submarine electronic support measures (ESMs) equipment	Trg	003
T0003	Maintain tactical plots	Trg	001
T0004	Maintain threat book	Trg	002
T0005	Maintain tactical logs	Trg	004
T0006	Operate submarine photographic equipment	Trg	004
T0007	Coordinate electronic warfare (EW) operations	Trg	003
T0008	Complete EW reports	Trg	003
T0009	Access the EW threat	Trg	003
T0010	Interpret EW display information	Trg	003
T0011	Analyse electronic intercept	Trg	003
T0012	Operate submarine friend or foe	No Trg	N/A
T0013	Set up digital camera	Trg	004
T0014	Apply camera corrections	Trg	004
T0015	Compose an image	Trg	004
T0016	Process digital imagery	Trg	004
T0017	Enhance a digital image	Trg	004
T0018	Catalogue images	Trg	004

Record of Proceedings

Dates: August 2019

Location: NTDC (A), Halifax, NS

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Inglis	P.E.	PO1	NCIOP	Submarine Support Team
Lacasse	J.	PO2	NCIOP	HMCS Chicoutimi
Neil	K.A.	PO2	NCIOP	HMCS Windsor
Lallemand	R.S.F.	PO2	NCIOP	HMCS Victoria
Bernard-Morissette	C.	MS	NCIOP	NTDC(A)
Chou	C.K.J.	MS	NCIOP	HMCS Victoria
Guthrie	N.A.	SLt	TDO	NTDC(A)

*Note: Some consultation done by correspondence.

Working Group Objective:

1. The objective of this working group is to reduce time away from home unit and improve the knowledge and confidence of students in the equipment used by integrating modern adult learning practices and methods of delivery.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communication checks	Taught on a prerequisite career course
A0010	Configure communications terminal equipment	Taught on a prerequisite career course
A0020	Operate C4I inter-ship circuits	Taught on a prerequisite career course
A0025	Operate CRYPTO equipment	Taught on a prerequisite career course
C0045	Operate combat system	Taught on CRWK(D) course
C0120	Calculate radar refraction prediction ranges	Taught on a prerequisite career course
C0125	Coordinate ship's/unit's request for information	Taught on RQ-MS course
C0150	Adhere to emission control restrictions	Taught on a prerequisite career course
C0160	Assist to the preparation of CD&E/OT&E/OPTEST test	Taught on a prerequisite career course
D0050	Operate multi TDL	Taught on a prerequisite career course
D0055	Manage multi tadil network	Taught on a prerequisite career course
D0065	Operate IFF	Taught on a prerequisite career course
D0070	Maintain recognized maritime picture	Taught on CRWK(D) course
D0115	Operate ECDIS	Taught on a prerequisite career course
K0009	Identification friend or foe	Taught on prerequisite course

K0010	Capabilities and limitations of masts and periscopes	Taught on prerequisite course
K0011	Submarine man aloft procedures	Taught on prerequisite course
T0012	Operate submarine friend or foe	Taught on a prerequisite career course

Discussions / Decisions:

a. It was agreed that this training shall use the modern training platform to enable training from home units using DLN platform, allowing more time at home and less time away from home.

b. The relevance of teaching IFF as IFF is being taught on previous courses.

c. The training was previously based on a specific electronic warfare system, fitted on Victoria class submarines. That system is in the process of being replaced. Part of the revision of this QSP addresses this by outlining training that is independent of specific equipment. The future employment of NCIOP onboard submarines will be done with new equipment, the QSP was thought to be required to cover both systems, thus allowing the students to be employed on all variants of Victoria class submarines.

d. The students graduating this course are to operate the system with minimal to no supervision, therefore they should be taught to perform at a level expected of a member of the team during OTT2

e. The small number of NCIOPs being loaded on this course can create delays in training. There should be pre-training allowing the trainee to be employable as part of the section in the event that their conversion course is delayed. The trainees should have access to trade specific training following their BSQ Mod 1, with a review package (PIP) prior to the course start. The DLN Module of this course is designed to be completed after BSQ Mod 1 to give students trade knowledge and allow them to be more employable. The PIP designed to be done prior to Module 2, ensuring students have the basic knowledge required to be successful on Mod 2, particularly if there has been an extended delay between Mod 1 and 2 or substantial time away from the submarine prior to Mod 2. In addition, the DLN can also be used by any NCIOP wishing to refresh on the equipment or theory.

f. The OJPR from the current AILD course is a great tool for the student and the unit to assess the operator's ability to perform their task in a submarine environment and should not be removed from course.

Guidance to Development and/or Instructional Staff:

a. All power points that do not require constant interaction between student and instructor, shall be replaced by DLN content to be instructed from home units using DLN/RAD.

b. IFF should be removed from this course.

c. The QSP shall not reflect a specific equipment, but rather be generic to the ESM suite as a generic term.

d. The training shall be in depth and challenging to the level of an OTT 2 watch or dived Work Ups. The course should focus on hands on performance based training rather than PowerPoint.

e. In order to maximize the time spent on course to hands on scenario, a pre-loading package could be used to complete the DLN portion of the course. This allows the trainee to have more tasks handed to them between their BSQ and the in-house training.

f. An OJPR should be used to ensure the trainee can perform their task in a real environment. This will allow the trainee to build confidence onboard and provide the home unit the ability to assess the level of supervision required for the trainee in a submarine environment.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	**Reassess the impact of shark fin requirements to the AILD QSP post pilot.**	**NTDC(A) with Sea Training Submarines**	**Arrival of Shark Fin +6 Months**
2	**Creation of video for PO 003 to allow ease of instruction and increased performance based learning.**	**NTDC(A) with MDA staff**	
3	**Creation of specific runs for PO 003 to better support the instructor during the trainer phase.**	**NTDC(A) with MDA staff**	

Costing Information

Include information about cost in whichever format is applicable (**Not to be deleted**).

Estimates only

Course: Sub NCIOP		Serial:			
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course
Airfare (Return)	6	$1,500.0	N/A	$8,000	$8,000
R&Q	6	$36.0	20	$4,320	4,320$
Incidentals	6	$17.5	20	$2,100	$2,100
Subtotal				**$18,420**	**$18,420**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)		
PP&S (Estimate)	6	$25.00	1	$150	$150
Computer Requirements (DLN home unit)	1 per student	$1,000.0	1	varies	varies
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0
DWAN Connection (Set up)	1	$5,000.0	1	$5,000	$0
Instructional Materials (Printed / Handouts etc)	6	$25.00	1	$150	$150
Subtotal				**$11,300.00**	**$300**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)		
Instructor (PO1/WO) (incremental)2	0	0	0	0	0
Military Guest Instructor	1	0	3	0	0
		subtotal		**$0**	**0**
	Total Estimated Costs			**$29,720**	**$18,720**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's competence in flash up and shut down of a submarine radar	DLN	Trainee must achieve 70% IAW EC 001.01 solution.
EC 001.02	EO 001.02	To assess student's competence in operating the submarine radar	Practical	Trainee must achieve 5 / 7 SATs IAW EC 001.02 checklist
EC 001.03	EO 001.03	To assess student's competence in performing a radar cut procedure	Practical	Trainee must achieve 3 / 5 SATs IAW EC 001.03 checklist
PC 001	PO 001	To assess student's competence in performing the duty of a radar watch keeper	Practical	Trainee must achieve all critical factors IAW PC 001 checklist
EC 002.01.01	EO 002.01.01	To assess student's competence in building the NCIOP portion of a mission brief	Practical	Trainee must achieve 40 / 56 SATs IAW EC 002.01.02 checklist
EC 002.01.02	EO 002.01.02	To assess student's competence in delivering the NCIOP portion of a mission brief	Practical	Trainee must achieve 5 / 7 SATs IAW EC 002.01.02 checklist
PC 002	PO 002	To assess student's competence in performing the intelligence duties onboard a submarine	Practical	Trainee must achieve 28 / 39 SATs IAW PC 002 checklist
EC 003.01	EO 003.01	To assess student's competence in flashing up EW equipment	DLN	Trainee must achieve 70% IAW EC 003.01 solution.
EC 003.02	EO 003.02	To assess student's competence on ESM reporting procedure	Practical	Trainee must achieve 70% IAW EC 003.02 solution.
PC 003	PO 003	To assess student's competence in operating ESM equipment	Practical	Trainee must achieve 11 / 18 SATs IAW PC 003 checklist

TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	LSR #
Interactive DLN	DLN	Increase interactivity of DLN	LG 001.01.01 and LG 003.01.01	
DLN Questions	DLN	Allow for quiz questions to be written in, prior to giving answers. Don't allow the answer to be revealed until the student tries to answer.	All DLN	

Scheduling Information

Scheduling Information – Submarine NCIOP AILD													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time (min)	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
Module 1													
Distance Learning (DL)													
001.01.01	50 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
002.01.01	50 est.	Various	0	0	0	0	Internet Access (DLN / Jane's)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
EC 002.01	400 est.	Various	0	0	0	0	Internet Access (DLN / Jane's)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
003.01.01	50 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
EC 003.01	50 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
003.02.01	50 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC
003.03.01	100 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	MIL	MS-PO2 NCIOP (subs)	SEC

Scheduling Information – Submarine NCIOP AILD													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time (min)	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
EC 003.03	50 est.	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	0	MS-PO2 NCIOP (subs)	SEC
004.02.01	100 est	Various	0	0	0	0	Internet Access (DLN)	UNC	0	0	0	MS-PO2 NCIOP (subs)	SEC
Module 2													
Pre-Instructional Package (PIP)													
Various	N/A	Home unit	0	0	0	0	N/A	N/A	0	0	0	MS-PO2 NCIOP (subs/from home unit)	SEC
Residential Phase													
001.02.01	400	Navigation Trg S-17	0	0	0	0	Trainer	UNC	0	0	Mil	MS-PO2 NCIOP (subs)	SEC
EC 001.02	300	Navigation Trg S-17	0	0	0	0	Trainer	UNC	0	0	Mil	MS-PO2 NCIOP (subs)	SEC
001.03.01	200	Navigation Trg S-17	0	0	0	0	Trainer	UNC	0	0	Mil	MS-PO2 NCIOP (subs)	SEC
EC 001.03	300	Navigation Trg S-17	0	0	0	0	Trainer	UNC	0	0	Mil	MS-PO2 NCIOP (subs)	SEC
PC 001	550	Navigation Trg S-17	0	0	0	0	Trainer	UNC	0	0	Mil	MS-PO2 NCIOP (subs)	SEC

Scheduling Information – Submarine NCIOP AILD													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time (min)	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
002.02.01	200	Classroom	1	0	0	1	N/A	UNC	1	6	Mil	MS-PO2 NCIOP (subs) /and EW1 qualified guest speaker	SEC
EC 002.02	150	Classroom /Briefing Room	1	0	0	1	N/A	UNC	1	9	Mil	MS-PO2 NCIOP (subs)	SEC
003.03.02	300	Fred Sherwood Trainer and Theater (S-17)	N/A	N/A	N/A	N/A	Trainer	SEC	N/A	N/A	Mil	MS-PO2 NCIOP (subs)	SEC
003.03.03	300	Fred Sherwood Trainer and Theater (S-17)	N/A	N/A	N/A	N/A	Trainer	SEC	N/A	N/A	Mil	MS-PO2 NCIOP (subs)	SEC
003.03.04	350	Fred Sherwood Trainer and Theater (S-17)	N/A	N/A	N/A	N/A	Trainer	SEC	N/A	N/A	Mil	MS-PO2 NCIOP (subs)	SEC

Scheduling Information – Submarine NCIOP AILD													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time (min)	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
003.03.05	2400	Fred Sherwood Trainer and Theater (S-17)	N/A	N/A	N/A	N/A	Trainer	SEC	N/A	N/A	Mil	MS-PO2 NCIOP (subs)	SEC
PC 003	900	Fred Sherwood Trainer and Theater (S-17)	N/A	N/A	N/A `	N/A	Trainer	SEC	N/A	N/A	Mil	MS-PO2 NCIOP (subs)	SEC
004.01.01	800	Formation Imaging	1	0	0	1	Nil	UNC	1	6	Mil	Qualified Imagery Tech	UNC

Plan, Train, Do

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01				Flash up Submarine Radar
	02				Track Contact
	03				Perform a Radar Cut Procedure
002	01		Build Mission Brief		
	02		Brief Mission Brief		
003	01				Flash up EW equipment
	02				Conduct ESM Reporting Procedures
	03				Operate ESM Equipment
004	01				Conduct Submarine Photography
	02				Conduct pre-sailing Duties

References

A = CF or other official Canadian government publication
B = Official allied force or allied government publication
C = Commercial publication
D = Other

Ref Code	Publication
A1	C-59-875-G00/MB-001 Colour Tactical Display (CTD3)
A2	CS-6700-14 AN/ULR 502 Sea search Operators Manual
A3	NCPM 401(B) – Victoria Class Combat Information Organization
A4	VCSOPs - Victoria Class Standard Operating Procedures
A5	A-PH-007-000/AG-001 CAF Unclassified Imagery Instructions

On Job Performance Record - Submarine NCIOP (AILD)

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

5. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be retained by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

6. As the trainee completes each Performance Factor (PF) to the required standard it will be signed and dated in their OJPR. The Training PO will review the OJPR weekly, or when reasonable progress has been made, and update Monitor Mass Training Tracker and inform the trainee of the action taken.

7. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO encounters any of these issues they are directed to contact NFSA Submarine Division for guidance.

PROGRESS REVIEW

8. Progress review shall be carried out weekly by the Section Head (EW1) and progress passed monthly to the COXN,

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

Section Head (EW1) Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Submarine Naval Combat Information Operator QSP except as noted has successfully completed on the job training. The member is qualified to carry out the duties of Submarine Naval Combat Information Operator and is AILD qualified.

Commanding Officer (CO) Date

I ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in this Annex shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

PERFORMANCE CHECKLIST

PO/EO	PERFORMANCE STATEMENT	DATE	SIGNATURE (EW1)	REMARKS
EO 001.01	Demonstrate radar flash-up procedure.			
EO 001.01	Demonstrate radar shutdown procedures.			
EO 001.03	Demonstrate radar cut procedures.			
EO 003.01	Demonstrate flash up/shut down/ reboot sea search II.			
EO 003.01	Demonstrate daily user check and confidence check of sea search II.			
EO 003.03	Demonstrate analysis and recording of a signal.			
EO 003.03	Demonstrate building and loading of a sea search library.			
EO 004.02	Describe the NCIOP set-up prior to departure to include: a. Bridge b. Control Room c. Warner Office			
PO 004	Set up control room to include: a. Wall paper b. FC c. Plot d. Warner e. Big Brother f. Little Sisters g. Camera(SRCH/ATT, Test Picture and Video)			
PO 004	Perform the duties of an NCIOP during a BINT to include: a. Picture recovery; b. BINT sheet; and c. Digital asset management.			
PO 004	Perform duties of submarine photographer to include: a. promotion/awards photos; b. evidence gathering photos; and			

	c. defect documentation.			
PO 002	Conduct a platform brief (PC 002)			

Item	PC 002 Conduct a Mission Brief	SAT	UNSAT	Remarks
Surface Platform #1				
Labeled correctly				
Picture is accurate				
All information on the card is filled in with accurate information to include: • Speed, Draught, Sec Ship, Safe Depth and GDR • Hull in Class • Helo • Radars • WPN				
Height Card #1				
Labels are clear				
Virtual height are in increase of 40 virtual				
All information is accurate to include: • Max speed • Draught • Second Ship • Go Deep Range				
Ranging heights are accurate including: • Mast Head • Side Lights • Stern Lights • Height between Lights				
Surface Platform #2				
Labeled correctly				
Picture is accurate				
All information on the card is filled in with accurate information to include: • Speed, draught, sec ship, safe depth and GDR • Hull in Class • Helo • Radars • WPN				
Height Card #2				

Labels are clear			
Virtual height are in increase of 40 virtual			
All information is accurate to include: • Max Speed • Draught • Second Ship • Go Deep Range			
Ranging heights are accurate including: • Mast Head • Side Lights • Stern Lights • Height between Lights			
Aircraft Platform #1			
Top row is labeled correctly			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Aircraft Platform #2			
Top row is labeled correctly			
Max speed is accurate			
Radar is accurate			
Range and endurance is accurate			
Indication of capabilities for night dipping, FLIR and MAD			
Picture is accurate			
Briefing Command			
The trainee was prepared			
The trainee displayed situation awareness			
Recommendation against Counter Detection			
Includes one recommendation against detection per platform			
Recommendation is pertinent			
Threat Matrix			
Includes all 10 possible encounters and all platforms from the brief			
Emitters are easily identified with the associated platform			
Matrix is broken down into section easy to read			

All radars are included			
Recognition Binder			
Easy to follow			
Key components are highlighted			
All platforms are included			
All platforms are properly labeled			
All pictures display proper marking (flag, hull number)			

Score: / 39

The trainee must achieve 28/39 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
EW1	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Pre-Instruction Package (PIP)

INSTRUCTIONS

1. The PIP must be completed prior to Module 2, but is not to be started more than 45 days prior to Module 2 start date.
2. The aim of the PIP is to review the information covered in Module 1 prior to the delivery of Module 2.
3. The student supervisor will email the instructor on completion of the PIP to confirm the review.

PO/EO	PERFORMANCE STATEMENT	DATE	SIGNATURE	REMARKS	MINIMUM QUALIFIQUATION FOR SIGNATURE
EO 001.01	Locate radar components to include: a. Transmitter/Receiver; b. Display (1007/JQA); c. Safe to TX Key; d. Safe to Rotate Key; e. Radar Mast; f. Wave Guide; g. Mast Desiccation; h. Mast Controls on the SCC; and Furuno				Any qualified submarine COMM TECH
AILS	Describe man aloft procedure, IAW current regulation.				DWS
AILS	Demonstrate man aloft procedures.				DWS
EO 001.01	Flash up radar.				EW1
EO 003.01	Locate sea search II components, to include: a. Antenna b. Antenna Auxiliary Power Supply c. Junction Box d. Receiver Processor Unit e. Microwave Tuner f. Data Processor Unit g. Operator Workstation h. Noise suppression enclosure				Any qualified submarine COMM TECH
EO 003.01	Go through pre-sailing check for sea search II.				EW1

EO 004.01	Prepare camera to include: a. Battery; b. Memory Card; c. Extra Batteries; and d. Memory Cards				Any qualified submarine NCIOP
EO 004.01	Set up attack periscope.				Any qualified submarine NCIOP
EO 004.01	Set up search periscope.				Any qualified submarine NCIOP